JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 93, NO. B3, PAGES 2191–2200, MARCH 10, 1988

Regional Magnetic Overprinting of Witwatersrand Supergroup Sediments, South Africa

PAUL W. LAYER[1,2], ALFRED KRÖNER[3], MICHAEL MCWILLIAMS[1], AND NORBERT CLAUER[4]

Shales from the late Archean Witwatersrand Supergroup of South Africa yield stable, well-grouped remanent magnetizations after thermal and alternating field demagnetization. Virtual geomagnetic poles from 17 sites located throughout the central and western parts of the basin indicate that rocks of the basin were remagnetized after folding. A paleomagnetic fold test shows that the Fisherian precision parameter, K, decreases from 22 to 3 after restoring the beds to paleohorizontal. Illite crystallinity indices show that the remagnetized samples were subjected to at least middle greenschist facies temperatures, whereas magnetic blocking temperature estimates indicate upper greenschist facies temperatures. A K-Ar age obtained from clays of 1945 ± 40 Ma (2σ) is assigned to this metamorphic event. The new paleomagnetic pole obtained from the Witwatersrand Basin sites (19°N, 46°E, $K = 22$, $A_{95} = 8°$) is consistent with other 1950–2050 Ma poles from Southern Africa, confirming independently the K-Ar age. The distinctive overprint direction seen in shale specimens may be useful as a directional marker for orienting deep bore cores.

INTRODUCTION

The Witwatersrand Basin located in South Africa contains some of the most thoroughly studied Archean gold deposits in the world; however, little is known about the thermal history of the basin. It is generally assumed that the basin was metamorphosed to no higher than lower greenschist facies [*Tankard et al.*, 1982]. Constraints on metamorphic temperatures are summarized by *Hallbauer and von Gehlen* [1983]. They argue that the maximum temperature reached should be constrained between the inversion temperature of marcasite (157°) and the decrepitation temperature of the secondary fluid inclusions in Witwatersrand quartz pebbles (250°C). Volatiles are expelled from carbonaceous material at temperatures above 250°C, so the apparently high volatile content in the Witwatersrand rocks also constrains this maximum temperature.

Oberthür and Saager [1986] report that this regional lower greenschist facies metamorphic event occurred about 2000 Ma. This metamorphic age is inferred from limited isotopic evidence [e.g., *Allsopp*, 1961] and (indirectly) from the supposition that the Bushveld Igneous Complex (BIC) is the most likely cause of the metamorphism. Until the present study, no direct evidence for the timing of this metamorphism has been published.

Recent evidence points to much higher metamorphic temperatures. *Phillips* [1987] has reported that mineral assemblages in Witwatersrand shales require temperatures in excess of 300°C and perhaps in excess of 400°C (upper greenschist facies).

A paleomagnetic study of this area should be able to help resolve this controversy by determining if the magnetization of the shales has been reset by a thermal event. A previous paleomagnetic study of the Witwatersrand Supergroup [*Henthorn*, 1972] was inconclusive because of the weak magnetic remanence in many samples and the fact that only a few sites were sampled, so a proper magnetic fold test could not be done.

In this paper we report results of a new, more thorough paleomagnetic study of samples from the Witwatersrand Supergroup and the Dominion Group. In addition, our study of illite crystallinity provides information about metamorphic grade, and K-Ar ages from the clay minerals are used to date this metamorphism.

GEOLOGIC SETTING

The late Archean to early Proterozoic history of the Kaapvaal Craton is highlighted by the development of the Witwatersrand Basin (Figure 1), a continuously expanding intracratonic basin which developed on Archean granite-greenstone basement in southern Africa, now partially exposed in the Johannesburg Dome [*Anhaeusser*, 1973]. The first strata deposited in the basin belong to the Dominion Group, followed in succession by the Witwatersrand Supergroup, the Ventersdorp Supergroup and the Transvaal Supergroup. The Witwatersrand and Transvaal strata are of major economic significance, due to the presence of gold, uranium and iron. These sequences contain sedimentary and volcanic rocks in varying proportions and cover a depositional interval from about 3000 to 2000 Ma [*Tankard et al.*, 1982]. Modern views consider the Witwatersrand strata as representative of a continental back arc foreland basin [e.g., *Burke et al.*, 1986].

The granitoid basement of central South Africa is composed of granites, gneisses and greenstones similar to the Archean basement exposed in the eastern Transvaal and in Swaziland [*Tankard et al.*, 1982]. In the central Kaapvaal Craton, the Johannesburg Dome is a multiphase pluton exposed in an ovoid window some 700 km^2 in extent (Figure 1). The oldest phase is a mixture of hornblende and biotite gneisses which are intruded by a younger (~ 3000 Ma) suite of homogeneous and porphyritic granodiorites [*Anhaeusser*, 1973]. The dome is onlapped by the lower units of the Witwatersrand Supergroup to the south, and by rocks of the Transvaal Supergroup to the north.

A five-point Rb-Sr isochron age of 3066 ± 65 Ma (recalculated with decay constants from *Steiger and Jäger* [1977])

[1]Department of Geophysics, Stanford University, California.
[2]Now at Department of Physics, University of Toronto, Canada.
[3]Institut für Geowissenschaften, Johannes Gutenburg-Universität, Mainz, West Germany.
[4]Centre de Sédimentologie et Géochemie de la Surface, Centre National de la Recherche Scientifique, Strasbourg, France.

Copyright 1988 by the American Geophysical Union.

Paper number 7B3020.
0148-0227/88/007B-3020$05.00

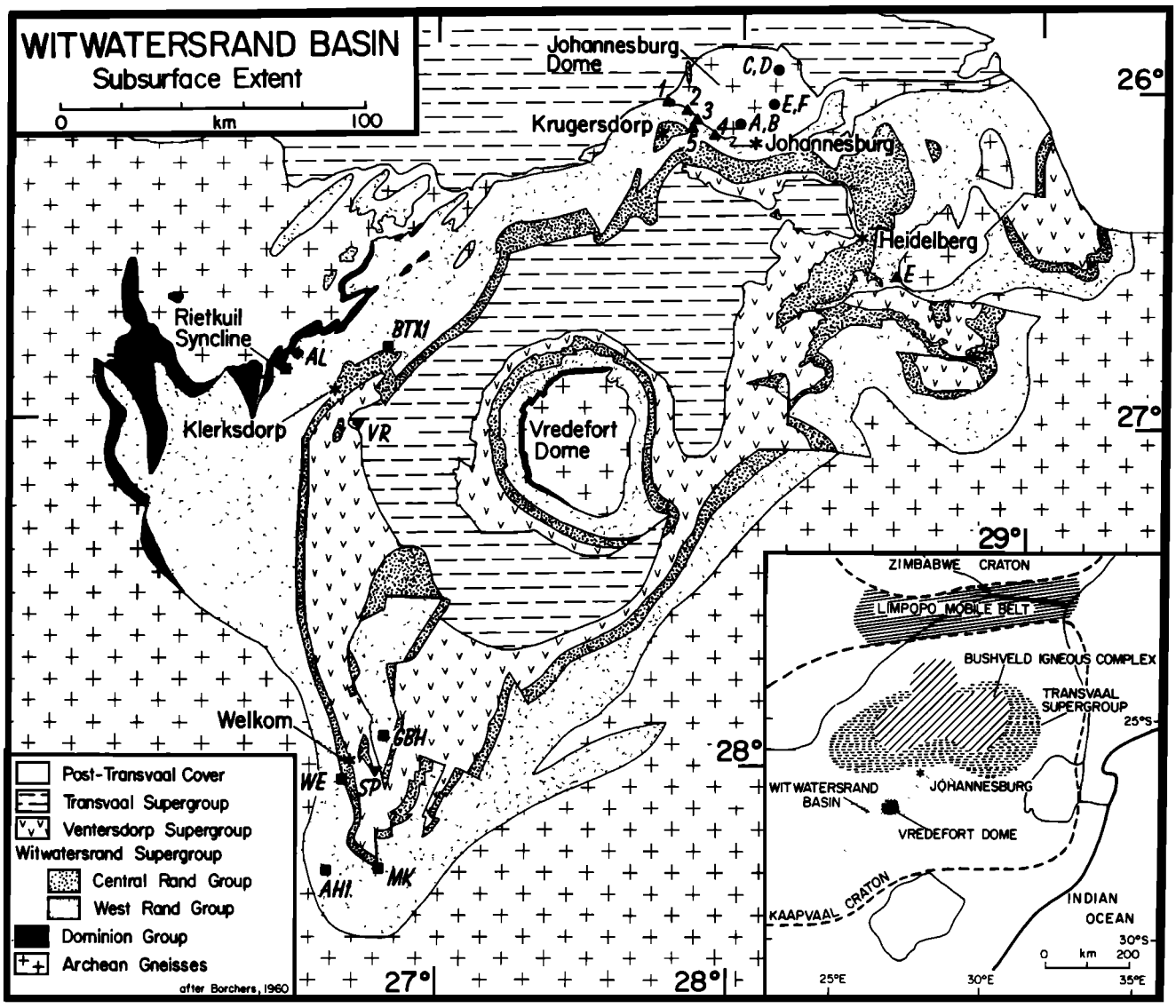


Fig. 1. Geologic map of central South Africa showing the subsurface extent of the Witwatersrand Basin in the central Kaapvaal Craton [after *Borchers*, 1960]. Localities sampled for paleomagnetic study are shown (see Table 1): triangle, Water Tower Shale (1–5, West Rand; E, East Rand); circle, Johannesburg Dome (A–F, sites AJA–AJF); diamond, Dominion Group; inverted triangle, Kimberley Shale; square, Jeppestown Shale. Inset is sketch map showing the location of late Archean and early Proterozoic features in southern Africa.

was found for the Johannesburg Dome by *Allsopp* [1961], and this age is believed to represent the upper limit for its formation. Allsopp found that some Rb-Sr mineral ages were younger than the overall whole rock isochron age, with the youngest mineral age being a biotite age of 2031 ± 10 Ma. He concluded that possible reheating by the nearby BIC might have reset the mineral ages.

Deposition of the Dominion Group reflects the protobasinal phase of the Witwatersrand Basin. The sediments and lavas of the Dominion Group are preserved over an area of about 15,000 km^2 with a maximum thickness of approximately 2710 m [*Tankard et al.*, 1982]. The basal sedimentary unit is composed of sandstone and conglomerate, all derived from underlying basement. The overlying and more predominant volcanic units vary in composition from basaltic andesite to acidic lava with quartz-feldspar porphyry. A single grain U-Pb zircon age of 3020 Ma has been obtained from one extrusive unit (R. Armstrong and W. Compston, Australian National University, personel communication, 1986).

The Dominion Group is overlain by the Witwatersrand Supergroup, covering about 100,000 km^2 with a maximum thickness of about 10,000 m in the deepest parts of the basin (Figure 1). The Witwatersrand Supergroup is subdivided into the lower West Rand Group, composed of the Hospital Hill, Government and Jeppestown subgroups, and the upper Central Rand Group, composed of the Johannesburg and Turffontein subgroups (Figure 2). The West Rand Group is finer grained and consists predominantely of quartzites and shales with rare conglomerate "reefs". The Central Rand Group is generally coarser grained and consists mainly of conglomerates and quartzites, with a single predominant shale horizon, the Kimberley (Booysens) Shale [*Eriksson et al.*, 1981]. Volcanic rocks are subordinate and are restricted to thin amygdaloidal flows of basaltic to dacitic composition [*Tankard et al.*, 1982].

The Witwatersrand Supergroup is overlain by rocks of the Ventersdorp Supergroup, which may represent the expression of intracratonal rifting [*Schweitzer and Kröner*, 1985;

21562202b, 1988, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB03p02191 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

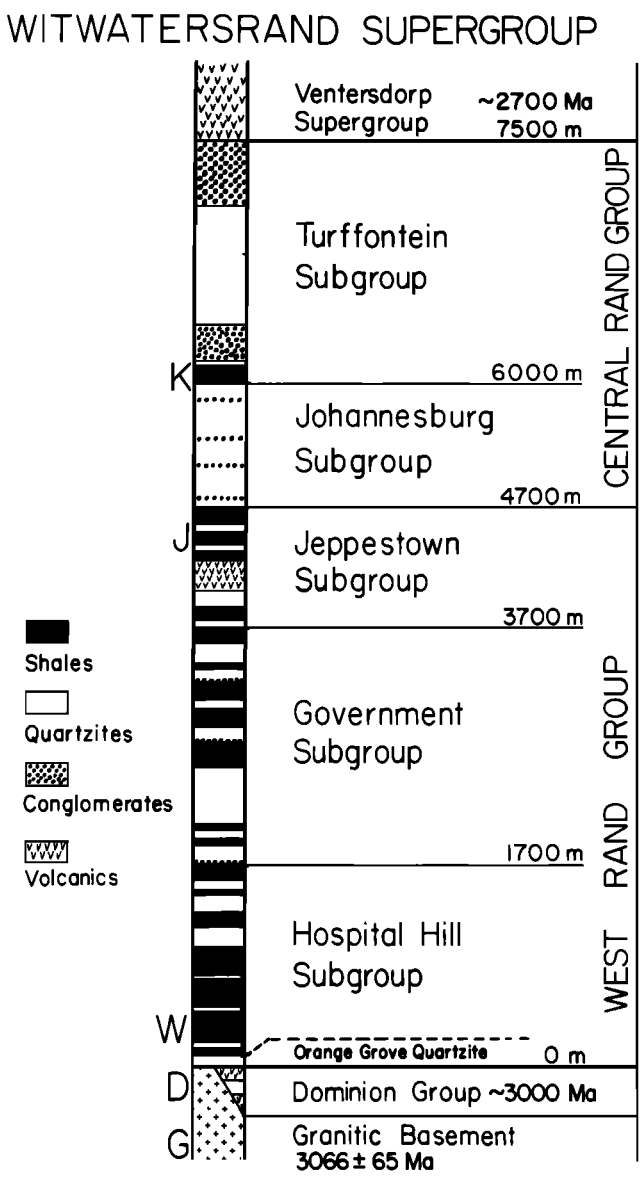


Fig. 2. Geologic column of the Witwatersrand Supergroup showing the major subgroups and their thicknesses. The various units sampled for paleomagnetic study are shown by single letter abbreviations (also used in Table 1): G, Johannesburg Dome granitoids; D, Dominion Group; W, Water Tower Shale; J, Jeppestown Shale; and K, Kimberley Shale.

Burke et al., 1986]. Ventersdorp strata overlie the Central Rand Group conformably at the center of the Witwatersrand Basin but define an angular unconformity at the northern margin of the basin, where the older rocks are block faulted and an angular discordance of about 5° exists. Large volumes of volcanic rocks with a generally bimodal composition were extruded over an area of at least 155,000 km^2 and reach a thickness of 5775 m. The intercalated sediments consist of conglomerates, shales, and some stromatolitic limestones. Feeders for some of the lavas appear as dikes and sills in the underlying Witwatersrand strata.

The age of the Ventersdorp Supergroup has been ambiguous for many years, but recent ion microprobe dating of single zircons has yielded a reliable concordia intercept age of 2699 ± 16 Ma [*Armstrong et al.*, 1986]. When coupled with the zircon age from the Dominion Group, this age constrains Witwatersrand Supergroup deposition between 3020 and 2699 Ma.

By Transvaal time (approximately 2200 Ma), the intracratonal basin had reached its greatest extent of about 500,000 km^2. Transvaal Supergroup rocks rest unconformably on older strata or transgress onto the granitoid basement. They reach a maximum thickness of 12,000 m and consist of carbonates, quartzites, minor volcanics, and banded iron formation. This association reflects deposition in a large epeiric basin [*Tankard et al.*, 1982].

At about 2000 Ma, intense igneous and thermal activity affected much of the central portion of the Kaapvaal Craton. The most significant manifestation of this activity is the BIC, considered to be the largest known ultramafic intrusion with a Rb-Sr whole rock isochron age of 2050 ± 12 Ma [*Hamilton*, 1977]. Many of the gabbroic dikes that cut the Witwatersrand and Ventersdorp sequences are probably related to the Bushveld event. The enigmatic Vredefort Dome, a circular structure southwest of Johannesburg, was uplifted at 1950 ± 60 Ma [*Hart et al.*, 1981] and might be related to the BIC intrusive event. In the vicinity of the Vredefort Dome, strata of the Dominion Group and the Witwatersrand, Ventersdorp, and Transvaal supergroups are deformed and overturned. There does not appear to be any evidence for major post-2000 Ma tectonic activity in this portion of the Kaapvaal Craton.

PALEOMAGNETIC STUDY

The Witwatersrand Supergroup is ideal for detailed paleomagnetic study because it spans approximately 300 m.y. and its stratigraphy is well documented (Figure 2). We originally anticipated that a late Archean magnetostratigraphy could be constructed from surface samples and oriented bore cores.

Sampling of surface outcrops was undertaken in the West Rand area near Johannesburg and in the East Rand area near Heidelberg (Figure 1 and Table 1) as part of a more regional paleomagnetic survey. Underground samples and oriented bore cores were obtained throughout the length of the Witwatersrand Basin from Johannesburg to Welkom.

Samples were taken at 20 surface outcrop sites, each site consisting of four to six oriented field-drilled cores. Cores were oriented by magnetic compass and (when possible) by a sun compass. To check the consistency of the magnetic direction obtained, two to four sites were usually located within 5 m (stratigraphically) and in similar lithologic units. These are here referred to as localities. Unweathered surface exposure of Witwatersrand Shale proved to be difficult to find, so only six localities were sampled, five in the West Rand and one in the East Rand.

Underground samples from five mines were collected as oriented blocks and occasionally as field-drilled cores. The underground samples were oriented by magnetic compass and with respect to the azimuth of adits taken from mine surveys. Sections of 13 oriented bore cores 15 – 50 cm in length were obtained from mining companies. These bore cores were either oriented by a downhole magnetic compass/camera system or by bedding cross sections with an assumed accuracy of ± 10°.

In the following analysis, each mine site or each bore core segment is considered to be a site, making a total of 38 sites when combined with the surface sites. This classification is correct for mine sites, where we have collected several independently oriented samples. With bore core sites, in only a

21562202b, 1988, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB03p02191 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1. Paleomagnetic Data

Site	Unit	N	Geographic D	Geographic I	Geographic Latitude	Geographic Longitude	κ	α_{95}	Pole	Stratigraphic D	Stratigraphic I	Stratigraphic Latitude	Stratigraphic Longitude
						West Rand Locality 1							
WOJ	W	4	357.6	−45.3	87.7	279.1	136	8	P	11.5	−23.7	72.6	68.2
WOK	W	4	307.9	59.3	8.0	350.2	215	6	a	273.2	45.2	−8.8	323.3
WOL	W	4	327.9	42.4	30.6	353.6	190	7	a	301.3	40.9	14.7	333.7
WOM	W	4	329.5	56.5	20.6	2.2	905	3	a	288.1	51.7	0.0	334.4
						West Rand Locality 2							
WON	W	4	31.4	59.1	17.8	52.6	20	21	A	203.1	79.9	−43.8	17.3
WOO	W	4	2.2	−58.4	76.9	200.3	13	26	P	17.7	−19.8	66.9	77.5
						West Rand Locality 3							
WOI	W	3	2.4	71.1	8.3	29.3	23	26	A	273.2	80.9	−23.8	8.5
WOS	W	4	26.4	63.3	15.1	47.0	211	6	A	19.0	84.3	−15.4	31.7
		2	208.0	38.0	−64.0	293.3			u	208.0	17.0	−58.2	269.6
WOT	W	4	38.2	70.6	2.7	48.8	62	12	A	148.3	86.3	−32.3	32.5
						West Rand Locality 4							
WOA	W	3	357.5	−39.9	85.8	351.6	140	10	P	6.2	−10.6	68.4	44.9
WOC	W	4	307.4	−44.5	43.1	285.5	11	20	u	337.5	−34.6	68.1	312.1
						West Rand Locality 5							
WOD	W	3	30.1	52.3	24.4	55.3	18	30	A	228.5	75.2	−41.8	359.8
WOE	W	3	32.4	71.3	3.5	45.3	92	10	A	218.7	53.6	−55.8	320.8
WOF	W	4	22.3	68.5	9.7	41.6	69	11	A	224.2	56.7	−51.1	325.6
WOG	W	4	345.7	−53.8	75.2	260.6	70	11	P	1.3	−3.1	65.4	30.9
						East Rand							
WOP	W	7	343.7	−55.1	73.4	261.6	12	19	P	19.7	−39.3	71.6	109.1
		3	352.1	61.3	20.6	22.3	30	24	A	296.2	52.2	5.1	339.3
WOQ	W	4	1.9	52.0	30.8	30.4	26	18	A	313.8	52.6	15.8	349.6
WOR	W	4	17.4	−43.2	74.3	117.1	17	23	P	31.5	−16.8	55.2	93.2
		5	49.9	60.8	7.5	63.6	17	19	A	297.3	79.2	−15.8	9.3
						Witwatersrand Basin Mines							
AL	D	1	18.3	41.0	37.0	47.6			A	339.3	56.7	23.3	8.7
VR	K	6	8.5	68.0	11.6	32.1	18	16	A	83.0	64.0	−14.4	72.4
		2	173.8	7.7	−66.1	191.3			u	175.0	−14.3	−55.4	197.9
SP	K	6	173.0	−2.5	−60.2	192.7	8	26	u	173.5	0.8	−61.9	193.0
WE	J	5	122.8	−70.4	5.8	177.5	1615	2	u	231.3	−73.1	6.4	231.0
		4	34.3	29.3	35.4	68.6	23	19	A	43.0	10.3	37.0	85.2
						Bore Cores							
AH1	J	4	305.6	80.0	−16.2	10.3	78	10	a	41.0	52.6	17.2	61.8
BTX1	J	3	17.8	60.8	19.4	40.8	82	14	A	91.2	50.6	−14.5	88.7
		3	33.6	67.4	7.4	47.7	182	9	A	119.6	51.6	−37.8	95.4
GBH1253	J	4	39.2	46.9	22.6	64.1	81	10	A	77.0	56.3	−6.9	78.7
GBH1263	J	4	286.3	−66.7	31.0	253.9	23	20	u	317.0	−43.8	51.7	290.0
		3	268.5	18.1	−5.6	304.4	51	17	u	257.3	22.3	−16.5	301.5
MK1	J	3	138.1	59.0	−58.3	87.2	828	4	u	166.0	83.0	−41.6	31.2
RKK25	J	4	9.5	32.2	44.8	39.3	45	14	A	359.5	18.2	53.9	25.7
RKK31	J	3	333.3	45.2	30.6	358.7	147	10	a	322.3	25.2	35.7	339.3
RKK35	J	2		53.0					a		53.0		
RKK38	J	3	325.4	49.7	23.8	354.2	103	12	a	329.6	68.6	6.9	8.2
						Johannesburg Dome							
JDA	G	3	40.0	39.3	28.0	70.4	11	39	a	40.0	39.3	28.0	70.4

Site, sampling sites, for mines (AL, Afrikander Lease; VR, Vaal Reefs; SP Saaiplaas Mine; WE, Welkom Mine); Unit, stratigraphic unit sampled (W, Water Tower Shale; D, Dominion Group; K, Kimberley Shale; J, Jeppestown Shale; G, Granite); *N*, number of specimens used in calculation of mean direction; Geographic *D*, *I*, Latitude and Longitude, site mean declination, inclination, latitude (°N), and longitude (°E) in uncorrected (in situ) coordinates; κ, Fisherian precision parameter; α_{95}, radius of circle of 95% confidence about mean direction; pole, paleomagnetic direction classification (P, present Earth field direction; A, direction A used in calculation of mean A pole; a, may be direction A, not used in calculation of mean A pole; u, unknown direction); Stratigraphic *D*, *I*, Latitude and Longitude; site mean declination, inclination, latitude (°N), and longitude (°E) after correcting bedding to horizontal. Sites WOS, WOP, WOR, VR, and WE have two magnetic vectors present in the same site. Sites BTX1 and GBH1263 have two different segments of core (treated as separate sites). No azimuthal information is availiable for site RKK35.

few cases do we have two or three separately oriented core pieces to check core orientation. In all other cases, three contiguous segments of core are designated a site. The spacing of samples in such a site is about 7 cm in stratigraphic section; all segments are referred to the same orientation.

The majority of samples come from shale units throughout the Witwatersrand Supergroup. In addition to these sites, one site in the Afrikander Lease Mine is located in strata of the Dominion Group, and six field-drilled sites are located in basement rocks of the Johannesburg Dome (Figures 1 and 2). Bedding at sampling sites shows a wide range of structural orientations.

Pilot specimens from selected sites were demagnetized in alternating fields (AF) up to 80 mT. Demagnetizations were performed along three perpendicular axes within the superconducting cryogenic magnetometer at Stanford University. Specimens from several sites were unstable during AF demagnetization; however, in the majority of cases, stable vec-

21562202b, 1988, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB03p02191 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

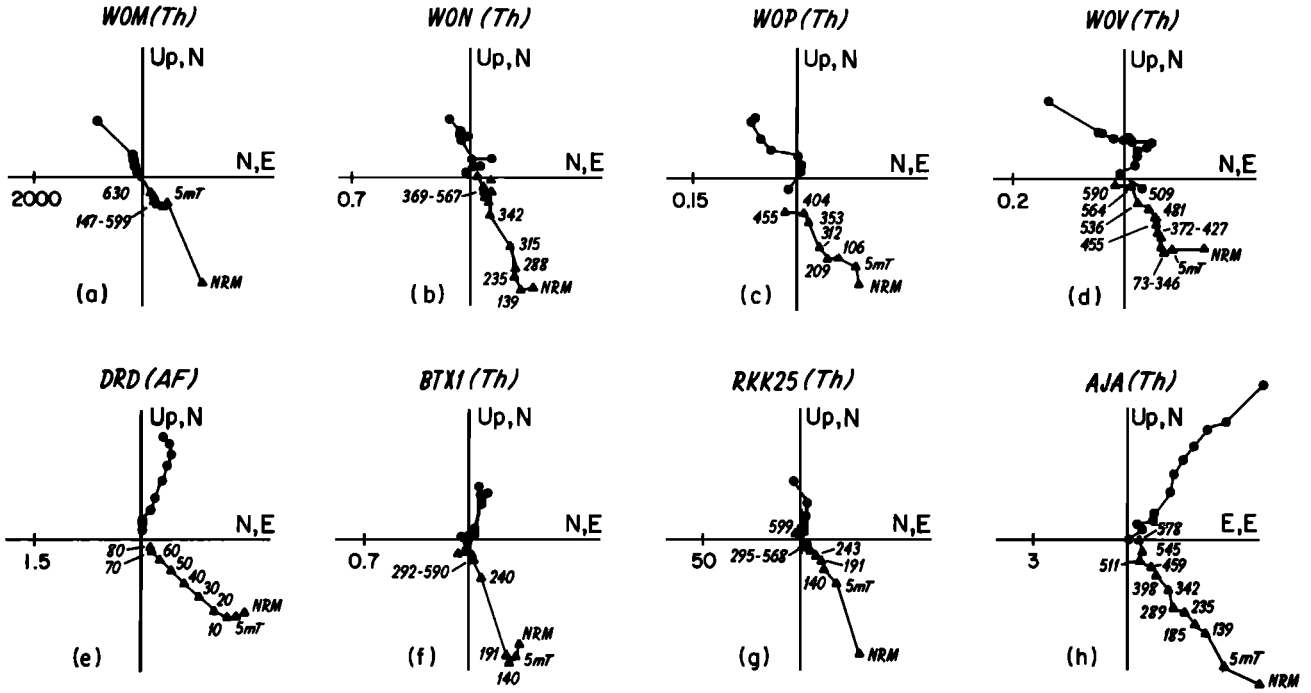


Fig. 3. Orthogonal vector plots of thermal (Th) or alternating field (AF) demagnetization of representative specimens from the Witwatersrand Basin area showing the A direction of magnetization. (*a*) and (*b*) West Rand Water Tower Shale samples, (*c*) East Rand Water Tower Shale, (*d*) Vaal Reefs mine Kimberley Shale, (*e*) Afrikander Lease mine Dominion Group lava, (*f*) and (*g*) Jeppestown Shales from bore cores, and (*h*) Johannesburg Dome granite. Circles show projection on the horizontal plane, and triangles show the projection on the vertical, north-south plane except for Figure 3*h*, which is on a vertical east-west plane. Scale values have been multiplied by 10^3 A/m.

tors could be isolated by stepwise thermal demagnetization in air up to 700°C using a Schonstedt TSD-1 furnace. Thermal demagnetizations were also done on samples from the sites with specimens that were stable during AF demagnetization in order to measure unblocking temperatures. Magnetization vectors were isolated using principal component analysis [*Kirschvink*, 1981].

Mean directions and associated uncertainties were obtained by use of Fisher statistics calculated on sample mean directions at each site. Site mean directions were calculated in both geographic (in situ) and stratigraphic (bedding corrected to paleohorizontal) reference frames. From these site mean directions, virtual geomagnetic poles (VGPs) were calculated for each site, assuming an axial geocentric dipole model. Site VGPs are averaged using Fisher statistics to obtain paleomagnetic poles and associated uncertainties for the Witwatersrand Supergroup both before and after restoration to paleohorizontal.

West Rand Area

We collected samples at five localities in the West Rand area north of Krugersdorp and Roodepoort (Figure 1). Four are from the Water Tower Shale (WTS) Member of the Hospital Hill Subgroup, while the fifth is from a green shale horizon above the WTS.

The first of the four WTS localities is 5 km north of central Krugersdorp and consists of 16 samples collected at four sites (WOJ, WOK, WOL, WOM). Most specimens have a large component of magnetization parallel to the present field direction which was removed by demagnetizing the specimen at 5 mT. Samples from site WOJ show a pronounced slaty cleavage, and specimens from these samples exhibit only a present field direction (P) during AF demagnetization. A stable magnetization different from the present field direction was isolated at the remaining sites (Figure 3*a* and Table 1). This component has a magnetic unblocking temperature of ~550°C. This direction (here referred to as direction A) was observed at most sites throughout the basin.

The second WTS locality is at Witpoortjie Falls, 7 km southeast of the first locality. Four samples at each of two sites (WON, WOO) were collected. Thermal demagnetization experiments on four specimens from site WON show the stable A direction (Figure 3*b*) with an unblocking temperature of 300°C. Site WOO is located in a somewhat weathered, shaly roadcut. All specimens from this site show the P direction with a much higher unblocking temperature (~650°C) than WON specimens. The magnetic carrier may be hematite, formed by oxidation of magnetite.

The third WTS locality is in Little Falls Gorge, 4.5 km southeast of WON and WOO. Samples were collected at three sites (WOA, WOB, WOC) in a cleaved green shale. All specimens from site WOA and one specimen from WOC show the P direction with a high coercivity. The remaining three specimens from site WOC yield a mean magnetic direction different from those of P or A. AF demagnetization experiments on specimens from site WOB show scattered directions.

The fourth locality is 3 km southeast of WOA, WOB, and WOC, a tuffaceous shale that shows a well-developed cleavage. Three sites (WOI, WOS, WOT) were collected. Step-

21562202b, 1988, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB03p02191 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

wise AF demagnetization reveals a stable magnetic component whose intensity decreased only to about 70% of the natural remanent magnetization (NRM) value when demagnetized in peak fields of 80 mT. This magnetization is similar to the A direction seen at other sites. Thermal demagnetization shows that this direction unblocks at 350°C, whereupon the intensity drops by a factor of 100 and a second, cleavage parallel, direction is seen in some specimens. The A direction appears to be independent of the orientation of cleavage in the samples when it is observed.

The fifth locality consists of five sites (WOD, WOE, WOF, WOG, WOH) and is located in Witpoortjie Gorge, 2.5 km upstream from sites WON and WOO. Although the exact stratigraphic position is not known, these samples come from a shale unit above the Water Tower Shale. At this locality the bedding dip is 55° as opposed to about 30° at the other localities. Site WOG is located in a massive green shale unit 100 m above the other four sites, and all specimens from it show the P magnetic direction. The other four sites were located in a green calcareous shale, and AF demagnetization results show the A direction of magnetization in three sites. Scatter at site WOH was too great for the mean direction to be considered significant.

East Rand Area

Fourteen samples were collected from the WTS at three sites in the East Rand area (WOP, WOQ, WOR). The beds here dip to the west, and results provide a fold test when combined with the West Rand data.

The shale occupies a small valley between two resistant quartzite units. Where water has penetrated the shale through joints, oxidation was visible. These specimens exhibited a P direction, seen in seven specimens from WOP and four from WOR. The magnetic carrier of this P overprint is probably hematite, formed by oxidation of magnetite. In most areas, the shale was not generally oxidized and appeared to be fresh. Stable A directions could be isolated in specimens at all three sites by both AF and thermal demagnetization (Figure 3*c*). Unblocking temperatures of 375°C were observed.

Witwatersrand Basin Mines

In three mines (Vaal Reefs Shaft 8, Saaiplaas Shaft 3 and Western Deep Levels), Kimberley Shale from the Turffontein Subgroup was collected. At Vaal Reefs, three oriented hand samples in the MB1 (Kimberley Shale equivalent) unit showed stable behavior during thermal demagnetization and an unblocking temperature of 450°C. A mean direction similar to the A direction seen in surface outcrops was calculated for this mine (Figure 3*d*).

Oriented hand samples from the Dagbreek Formation in the Saaiplaas Mine, showed a mean remanence direction different from that seen in Vaal Reefs or elsewhere in the basin. The direction seen in the Saaiplaas Mine is parallel to bedding and cleavage and may reflect magnetic anisotropy in these highly deformed units. In the Western Deep Levels Mine, four oriented drill cores from five sites in the Kimberley Shale showed no stable magnetic components upon either AF or thermal demagnetization.

Five oriented hand samples of the upper Jeppestown Shale were collected in the Welkom Shaft 4. Two magnetizations were recognizable in demagnetization experiments. The first component was characterized by a high median destructive field and an unblocking temperature of about 340°C. This component is not seen at other sites in the basin and the remanence carrier is not known. The underlying component has an unblocking temperature of about 500°C and is seen in four of the five samples. It has an average intensity 100 times less than the first component and is an A direction.

In the Afrikander Lease Mine, six oriented basalt and quartzite samples were collected from the Dominion Group. The three quartzite samples were magnetically weak and did not show a coherent direction. One of the three basalt samples (DRD) has several specimens which exhibited stable single component behavior during both AF and thermal demagnetization experiments and has an A direction component (Figure 3*e*). The other two basalt samples showed slight alteration and were unstable during demagnetization.

Bore Cores

Oriented bore cores from the central and southern Witwatersrand Basin were obtained from Anglo-American Corporation and Anglo Transvaal Consolidated Investments Company. With two exceptions, one section of core was obtained per hole. Core BTX1 yielded two sections 40 m apart (in depth), and GBH1263 yielded two sections 20 m apart (in depth). Stable magnetizations were seen in shales of the Jeppestown Subgroup (Figure 3*f*), which vary in composition from medium-grained gray quartz-rich shale to fine-grained green shale. Unblocking temperatures from 350° to 500°C are observed from these cores.

Five oriented bore cores were obtained from the Rietkuil Formation (part of the Jeppestown Shale in the Rietkuil Syncline near Klerksdorp) for a local fold test study. All cores were vertically drilled, and the azimuthal orientation for paleomagnetic measurement was determined by aligning the bedding in the core with field measurements. No bedding information was available for core RKK35, but the bedding in this core was horizontal. We assumed that the inclination before and after restoration of bedding to paleohorizontal did not change.

Using thermal demagnetization, we were able to isolate a stable magnetization in four of the five bore cores (Table 1). These cores show only the A magnetization seen in surface and mine sites (Figure 3*g*). The fifth bore core had a intense drill-string overprint: a bore core plunge-parallel magnetic direction induced by remagnetization during drilling. Any underlying magnetization could not be isolated with confidence. Using the three stable A bore cores, for which both declination and inclination of the paleomagnetic direction could be determined, the fold test is clearly negative, with κ decreasing from 16 to 6 upon correction to paleohorizontal (Figure 4). Some dispersion in declination may have been produced by orientation error, because the core was oriented using bedding attitudes only. If we consider only inclination and therefore include RKK35, values are much less scattered before restoration to paleohorizontal than afterward. The in situ mean inclination from the four stable cores is similar to that of direction A seen throughout the basin.

Johannesburg Dome

Twenty-four samples were collected at six sites distributed around the Johannesburg Dome. Sites AJA and AJB were located in a stream floor in a medium- to coarse-grained pinkish gray granodiorite near Johannesburg. Sites AJC and AJD come from a migmatitic banded gneiss in the Tarmac

21562202b, 1988, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB03p02191 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

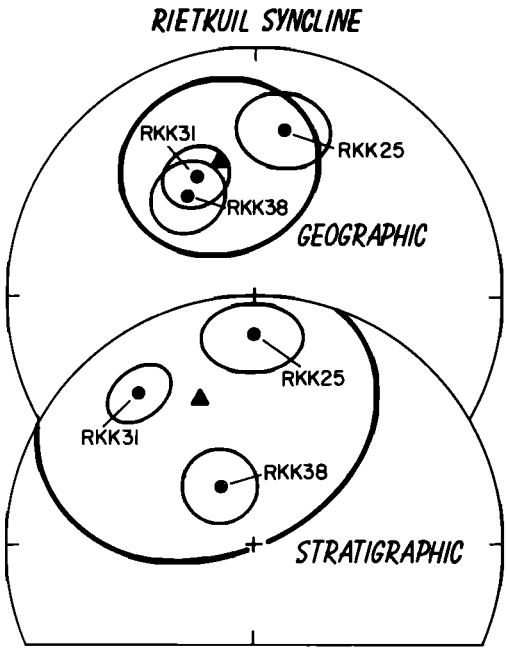


Fig. 4. Equal area stereographic projections of the results from the Rietkuil Syncline before (geographic) and after (stratigraphic) bedding correction is applied. Solid circle is an individual bore core mean direction as plotted on the lower hemisphere with its associated α_{95} plotted as a small circle about the mean. Large triangle is the mean value calculated for the three bore cores along with its α_{95} plotted as a small circle.

Ross Quarry 30 km north of Johannesburg. Sites AJE and AJF are gneiss samples from the Halfway House Quarry 17 km north of Johannesburg.

Specimens from all sites except AJA and AJB showed erratic, unstable behavior upon demagnetization. These samples show stable single-component demagnetization vectors, with a magnetic carrier displaying a range of unblocking temperatures between 100°C and 570°C (Figure 3*h*). This magnetization also has a low median destructive field (about 7 mT) during AF demagnetization, suggestive of multidomain magnetite as the primary remanence carrier. The specimens demagnetized thermally yielded a mean direction similar to the A direction from the in situ Witwatersrand sites.

SUMMARY OF PALEOMAGNETIC RESULTS

Results from all sites in the Witwatersrand Basin are listed in Table 1. Component A is either the only component of magnetization or the higher unblocking temperature component at 23 sites. An unblocking temperature range of 400 ± 100°C is observed for this component. Six sites show direction P as the only component or as the lower unblocking temperature component.

The paleomagnetic pole for component P calculated from site mean VGPs is summarized in Table 2. The confidence limit for P includes both the present magnetic (dip) pole and the present geomagnetic pole and thus probably represents a recent overprint due to weathering. The P magnetization was observed only at surface exposures, thus supporting this conclusion.

Figure 5 shows site mean VGPs associated with the A direction in the geographic (in situ) and stratigraphic (bedding corrected) reference frames. The site mean directions from which the VGPs were calculated are of uniform polarity. Seventeen of the 23 VGPs lie in a distinctive cluster near northeast Africa in geographic coordinates. The remaining six poles seem to lie 15°– 30° west of the main group. Three of these discordant VGPs are from the West Rand locality 1, the farthest west of the surface localities. Here the strike of the bedding is also rotated 15°– 30° counterclockwise compared with other West Rand localities. T. McCarthy (University of Witwatersrand, personel communication, 1984) has postulated that this area was subjected to later faulting, and our data indicate that this occurred following acquisition of the A direction. Two VGPs are from the subsurface core in the Rietkuil Syncline. These cores have declination uncertainties of ±10°, which may account for the apparent rotation of the A direction. The Rietkuil Syncline also lies near the western margin of the basin, and it is possible that it was rotated at the same time as the West Rand locality 1, although there is no supporting evidence for this model. The other VGP is from core AH1, and it is possible that it was misoriented. The mean paleomagnetic pole calculated for these six discordant sites is significantly different at the 95% limit from the pole for the remaining 17 (in situ), and these VGPs are neither included in the final Witwatersrand paleomagnetic pole, nor the paleomagnetic fold test.

From Table 2 and Figure 5 it can be seen that the precision, K, decreases from 22 to 3 upon tectonic correction, indicating that the A magnetization was acquired after these sites were tilted to their present attitude. Application of an incremental fold test shows that 0% tectonic correction yields the maximum K value. If the 6 rotated site VGPs are included in the population, the 23 VGP fold test also fails at the 99% confidence limit [*McElhinny*, 1964], with

TABLE 2. Paleomagnetic Poles

Study	Age, Ma	N	Latitude,°N	Longitude,°E	A_{95}	K	Reference
Bushveld Complex	2050 ± 12	63	16	30	4	21	*Hattingh* [1983]
Vredefort Ring	1950 ± 60	10	28	34	11	21	*Hargraves* [1970]
Limpopo Mobile Belt							
Direction A	2000	21	26	23	10	23	*Morgan* [1985]
Lower Ventersdorp	2500	4	-4	40	11	8	*Henthorn* [1972]
Witwatersrand Supergroup							
Direction P	PEF	6	85	232	10	43	this study
Direction A geographic	1945 ± 40	17	19	46	8	22	this study
Direction A stratigraphic		17	-18	28	26	3	this study

Study, paleomagnetic study; Age, best estimate for the age of magnetization (PEF, present earth field); N, number of sites; Latitude, Longitude, location of the paleomagnetic pole; A_{95}, radius of circle of 95% confidence about the mean pole; K, Fisherian precision parameter; Reference, literature reference for the pole.

21562202b, 1988, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB03p02191 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

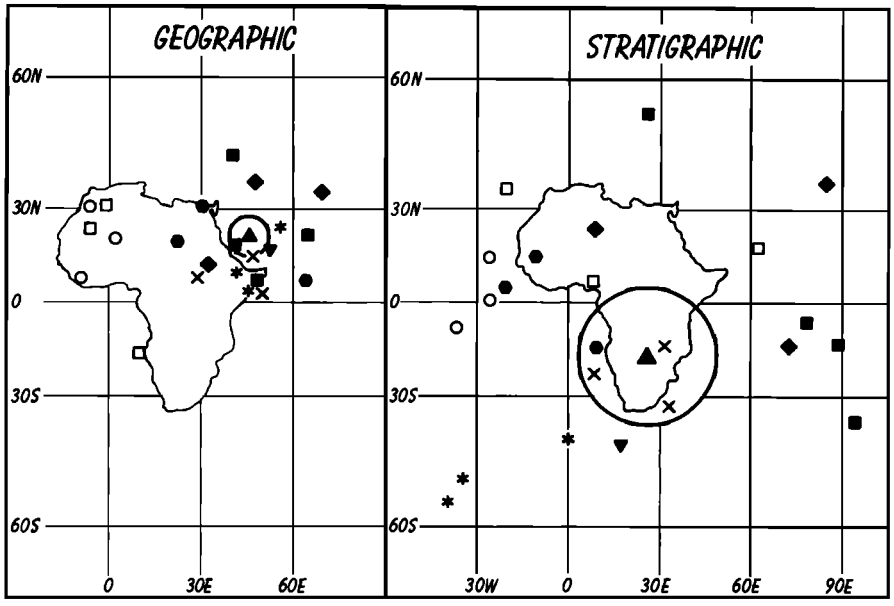


Fig. 5. Mercator map showing the site means (small circles) for the 23 sites which have an A component of magnetization in geographic (in situ) coordinates and stratigraphic (bedding corrected) coordinates. Open circle, West Rand locality 1; inverted triangle, locality 2; cross, locality 3; star, locality 5; hexagon, East Rand; diamond, mines; solid and open squares, bore cores. Open circles and squares are site mean VGPs not included in the calculation of the mean pole. Site A_{95} circles have been omitted for clarity. Large triangle shows the mean of the 17 solid symbols and is shown with the A_{95} for this paleomagnetic pole.

K decreasing from 10 to 3; however, for reasons previously stated, we prefer to omit those six sites from the calculation of the paleomagnetic pole.

Eight sites yielded directions which differ from either the A or P mean direction. The majority of these anomalous directions have been previously discussed and may be due to anisotropy or drill-string overprinting, as some magnetic directions are parallel to borehole orientation. Bore core site MK1 was oriented by a scratch line cut into the core. We know from drilling logs that it is possible that this is in error by 90°. Because the core was vertically drilled, the inclination is similar to that of the average A direction, but the declination is not in agreement, supporting the misorientation idea.

The most stable magnetization from the Johannesburg Dome is indistinguishable from the A direction within experimental uncertainty. This agreement is due in part to the large A_{95} associated with the Johannesburg Dome VGP. However, if the Witwatersrand Basin was overprinted over its entire length subsequent to basin formation, it is likely that the Johannesburg Dome was remagnetized at the same time. Because only one of the six sites sampled showed stable, consistent results, it appears that the Johannesburg Dome is not a good candidate for further paleomagnetic study. Due to the poor quality of the results from the Johannesburg Dome, the site AJA VGP is not included in the overall calculation of the Witwatersrand paleomagnetic pole.

It appears, then, that the A direction of magnetization is the one most representative of the Witwatersrand Basin, and it was acquired after the basin had achieved its present shape. Either new magnetic minerals formed in the shales or previously existing magnetic minerals were reset at that time. Using the model of *Pullaiah et al.* [1975], a temperature of 425°C is needed for 1 m.y. to reset the magnetization with the range of unblocking temperatures seen in the laboratory for component A. This temperature is very close to new geothermometry data of *Phillips* [1987]; however, it is much higher than previous estimates [*Tankard et al.*, 1982; *Hallbauer and von Gehlen*, 1983]. In some shale specimens, euhedral magnetite grains are seen. These may be secondary, formed at the time of metamorphism; however, they are not seen in all specimens showing the A direction. We propose that, in general, magnetic grains were reset in the shale samples at upper greenschist facies temperatures, and at some locations in the basin, conditions allowed for growth of secondary magnetite. In both cases, the specimens from the shales show the A direction.

CLAY MINERALOGY AND K-AR AGES

Our suspicion of widespread remagnetization and the presence of micas in hand specimens led us to investigate their clay mineralogy and crystallinity. We wished to test previous claims that the Witwatersrand strata were subjected to peak temperatures considerably lower than those implied by the results of our paleomagnetic study.

Clay fractions smaller than 2μm were extracted from six samples of the cleaved shales from the West Rand area using a sedimentation technique in distilled water. The X-ray diffraction data are summarized in Table 3 and give estimates of the major clay mineral fractions as well as the crystallinity index. The major component of all clay fractions is chlorite, ranging from 50% to 100%. Chlorite is mixed with illite and traces of quartz and feldspar, and in three samples, traces of carbonates are found. The illite crystallinity index averages 2.2 ± 0.2, suggesting recrystallization of the original rock under greenschist facies conditions. The polymorphic form of illite is 2M, which, in combination with the

21562202b, 1988, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB03p02191 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3. Clay Mineralogy

Sample	Illite %	Chlorite %	Index†	Other Minerals‡
WOA	50	50	2.0	Qz*,F,Ca
WOB	50	50	2.0	Qz*,F,Ca
WOC	30	70	2.4	Qz,F,Ca
WOE	0	100		Qz*
WOF	25	75	2.0	Qz
WOH	20	80	2.1	Qz

†Crystallinity index after *Kubler* [1967], limit of zeolite facies - greenschist facies at 3.5.

‡Qz, quartz; F, feldspars; Ca, carbonates; *, large amount.

TABLE 4. K-Ar Data

Sample Name	K_2O wt %	Ar* %	$^{40}Ar^*$ $\times 10^{-6}$ cm^3 STP/g	Age Ma $\pm 2\sigma$
WOB	4.53	96.75	48.87	1893 ± 45
WOF	2.07	95.81	23.87	1972 ± 64
WOH	3.26	98.61	37.57	1971 ± 53

$^{40}K/K_{total} = 1.167 \times 10^{-4}$ mol/mol; decay constants for ^{40}K are $\lambda_{\beta^-} = 4.962 \times 10^{-10}$ yr^{-1}; $\lambda_e + \lambda_{e'} = 0.581 \times 10^{-10}$ yr^{-1} [*Steiger and Jäger*, 1977].

crystallinity, suggests that a thermal event of at least 250°C is recorded by the clay minerals.

K-Ar dating was performed on clay fractions smaller than 2μm from three samples, using conventional techniques. The three fractions give a consistent age of 1945 ± 40 Ma (2σ) (Table 4), which can be considered as the age of metamorphic recrystallization [*Clauer*, 1979, 1981; *Bonhomme*, 1982]. From the excellent agreement of this age and the age of emplacement of the BIC and the temperatures implied for metamorphism, we conclude that (1) recrystallization and paleomagnetic overprinting were effectively synchronous, and (2) resetting must be associated with a distinct metamorphic event of at least greenschist facies of considerable regional extent.

DISCUSSION

Figure 6 and Table 2 compare the paleomagnetic pole for the Witwatersrand Supergroup (WS) with other late Archean to early Proterozoic paleomagnetic poles from the BIC [*Hattingh*, 1983], Vredefort Dome (VR) [*Hargraves*, 1970], the Limpopo Mobile Belt overprint (LBA) [*Morgan*, 1985] and the Lower Ventersdorp (LV) [*Henthorn*, 1972]. It can be seen that our new results (in geographic coordinates) are consistent with most of the ~2000 Ma paleomagnetic poles, but do not agree with the result from the Lower Ventersdorp. This latter pole is calculated from 25 samples collected at two localities in the center of the Witwatersrand Basin (Western Deep Levels Mine and President Styen Mine), after restoration of bedding to paleohorizontal. The result is in direct conflict with our Witwatersrand results, because in arriving at it, the author assumed that the magnetization at these sites was acquired prior to folding. We believe that these samples are not fully representative of the magnetization of the Lower Ventersdorp as a whole, and that the two-locality fold test is not conclusive, so the Lower Ventersdorp pole should not be considered to be a valid Precambrian paleomagnetic pole for southern Africa. Our study is based on 17 stratigraphically diverse sites with a much wider geographic distribution, and the results conclusively fail the fold test. In addition to the basinwide negative fold test, local evidence from the Rietkuil Syncline suggests that magnetization was acquired after formation of much smaller structures. However, some later, small-scale rotations may have occurred near the basin margin following remagnetization.

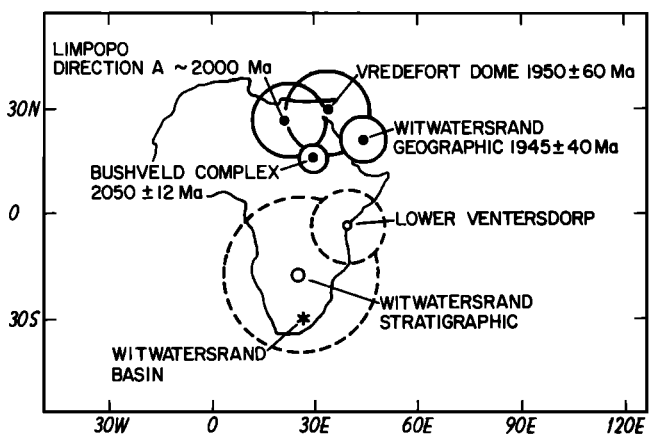


Fig. 6. Mercator map showing the 1950–2050 Ma paleomagnetic poles for southern Africa as solid dots with solid A_{95} confidence circles: Bushveld Igneous Complex [*Hattingh*, 1983]; Vredefort Dome [*Hargraves*, 1970]; Limpopo Mobile Belt direction A [*Morgan*, 1985]; Witwatersrand Supergroup without bedding correction (this study). The Lower Ventersdorp [*Henthorn*, 1972] and Witwatersrand Supergroup (this study) poles calculated after restoring the bedding to paleohorizontal are also shown as open circles with dashed A_{95} confidence circles.

The four paleomagnetic poles in agreement (WS, LBA, BIC, VR) are from units forming a north-south trend in the central Kaapvaal Craton from the Limpopo River in the north to Welkom in the south. At the center of this area is the BIC. We propose that the thermal aureole around the BIC as well as the regional thermal events that followed the intrusion of the BIC affected much of the central Kaapvaal Craton and remagnetized preexisting units such as the Witwatersrand Basin fill. At the same time, the Limpopo Mobile Belt was apparently uplifted and cooled [*Morgan*, 1985]. The VGP precision parameters, K, for these four poles are quite similar (20–22) and may reflect some intrinsic feature of the geomagnetic field at the time of aquisition of magnetization.

Older units to the west and east of this trend do not show the same degree of remagnetization. Poles from the Modipe Gabbro [*Evans and McElhinny*, 1966] and the Gaborone Granite [*Evans*, 1967] in Botswana as well as those from the Komati Formation in eastern South Africa [*Hale and Dunlop*, 1984] and the Usushwana Complex in Swaziland [*Layer et al.*, 1988] are associated with pre-2000 Ma cooling ages. Thus the effect of the Bushveld intrusion and cooling was more strongly felt through the center of the craton than at its edges.

It is possible that the Witwatersrand sediments were remagnetized after BIC intrusion, during the event that created the Vredefort Dome. The K-Ar age of the shales (1945 ± 40 Ma) is identical to that of the Vredefort Dome (1950 ± 60 Ma [*Hart et al.*, 1981]), however, it is significantly younger than the BIC age of 2050 ± 12 Ma [*Hamilton*, 1977]. In this case, then, the centrally located Vredefort Dome was formed by the same event that remagnetized and metamorphosed the Witwatersrand Basin. Currently, the nature of such an event is not fully understood, and it is not known

21562202b, 1988, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB03p02191 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

if it could have had regional thermal effects on the Witwatersrand Basin.

Our paleomagnetic and clay mineralogy studies provide independent confirmation of the hypothesis that the Witwatersrand Basin was reheated to middle or upper greenschist facies temperature at approximately 1950–2050 Ma. This is the first quantitative evidence that the entire central Witwatersrand Basin, from Johannesburg to Welkom, may have been thermally affected by the BIC (or, perhaps, the Vredefort event).

Although the remagnetization overprint is so pervasive that it prevents the determination of a primary magnetostratigraphy, the magnetization may be a useful means by which to evaluate the orientation of bore core material. The major exploration tool used in the Witwatersrand Basin is deep drilling. Most exploratory cores are not oriented, but the unique paleomagnetic signature of the overprint can provide an independent determination of subsurface core orientation, which can then be used to determine subsurface structure such as bedding orientation. Simple paleomagnetic surveys of fine-grained horizons in the core could be used to determine core orientation and thereby subsurface structure in areas of new drilling and exploration.

Acknowledgments. We thank Anglo-American Corporation, Anglo-Transvaal Consolidated Investment Company, and Johannesburg Consolidated Investment Company for field assistance and access to mines and bore cores. This research was supported by National Science Foundation grant EAR83-13482 to M.M. and by grant Kr 590/11 from the West German Research Council (DFG) to A.K.

REFERENCES

Allsopp, H. L., Rb-Sr age measurements on total rock and separated mineral fractions from the Old Granite of the central Transvaal, *J. Geophys. Res., 66,* 1499–1508, 1961.

Anhaeusser, C. R., The geology and geochemistry of the Archean granites and gneisses of the Johannesburg-Pretoria dome, in *Symposium on Granites, Gneisses and Related Rocks,* edited by L. A. Lister, pp. 361–386, Geolgical Society of South Africa, Johannesburg, 1973.

Armstrong, R. A., W. Compston, E. A. Retief, and H. J. Welke, Ages and isotopic evolution of the Ventersdorp Volcanics (abstract), in *Geocongress'86, An International Earth Science Congress,* pp. 89–93, Geological Society of South Africa, Johannesburg, 1986.

Bonhomme, M. G., The use of Rb-Sr and K-Ar dating methods as a stratigraphic tool applied to sedimentary rocks and minerals, *Precambrian Res; 18,* 5–25, 1982.

Borchers, R., Exploration of the Witwatersrand System and its extensions, in *The Geology of Some Ore Deposits in Southern Africa, vol. 1,* edited by S. H. Haughton, pp. 1–24, Geological Society of South Africa, Johannesburg, 1960.

Burke, K., W. S. F. Kidd, and T. M. Kusky, Archean foreland basin tectonics in the Witwatersrand, South Africa, *Tectonics, 5,* 439–456, 1986.

Clauer, N., A new approach to Rb-Sr dating of sedimentary rocks, in *Lectures in Isotope Geology,* edited by E. Jäger and J. C. Hunziker, pp. 30–51, Springer-Verlag, New York, 1979.

Clauer, N., Rb-Sr and K-Ar dating of Precambrian clays and glauconites, *Precambrian Res., 15,* 331–352, 1981.

Eriksson, K. A., B. R. Turner, and R. G. Vos, Evidence of tidal processes from the lower part of the Witwatersrand Supergroup, South Africa, *Sediment. Geol., 29,* 300–325, 1981.

Evans, M. E., A paleomagnetic study of the Gaberones granite of Botswana, *Geophys. J. R. Astron. Soc., 12,* 491–498, 1967.

Evans, M. E., and M. W. McElhinny, The paleomagnetism of the Modipe Gabbro, *J. Geophys. Res., 71,* 6053–6063, 1966.

Hale, C. J., and D. J. Dunlop, Evidence for an early Archean geomagnetic field: A paleomagnetic study of the Komati Formation, Barberton Greenstone Belt, South Africa, *Geophys. Res. Lett., 11,* 97–100, 1984.

Hallbauer, D. K., and K. von Gehlen, The Witwatersrand pyrites and metamorphism, *Mineral. Mag. 47,* 473–479, 1983.

Hamilton, P. J., Sr isotope and trace element studies of the Great Dyke and Bushveld Mafic Phase and their relation to early Proterozoic magma genesis in southern Africa, *J. Petrol., 18,* 24–52, 1977.

Hargraves, R. B., Paleomagnetic evidence relevant to the origin of the Vredefort Ring, *J. Geol., 78,* 253–263, 1970.

Hart, R. J., H. J. Welke, and L. O. Nicolaysen, Geochronology of the deep profile through Archean basement at Vredefort, with implications for early crustal evolution, *J. Geophys. Res., 86,* 10,663–10,680, 1981.

Hattingh, P. J., A paleomagnetic investigation of the layered mafic sequence of the Bushveld Complex, Ph.D. thesis, Univ. of Pretoria, Pretoria, South Africa, 1983.

Henthorn, D. I., Paleomagnetism of the Witwatersrand Triad Republic of South Africa and related topics, Ph.D. thesis, Univ. of Leeds, Leeds, England, 1972.

Kirschvink, J. L., The least squares line and plane and the analysis of paleomagnetic data, *Geophys. J. R. Astron. Soc., 62,* 699–718, 1981.

Kubler, B., La cristallinite de l'illite et les zones tout a fait superieures du metamorphisme, *Colloque sur les etages tectoniques,* pp. 105–122, Univ. Neuchatel, A la Baconniere, Neuchatel, 1967.

Layer, P. W., A. Kröner, M. McWilliams, and A. Burghele, Paleomagnetism and age of the Archean Usushwana Complex, southern Africa, *J. Geophys. Res., 93,*, 449–457, 1988.

McElhinny, M. W., Statistical significance of the fold test in paleomagnetism, *Geophys. J. R. Astron. Soc., 8,* 338–340, 1964.

Morgan, G. E., The paleomagnetism and cooling history of metamorphic and igneous rocks from the Limpopo Mobile Belt, southern Africa, *Geol. Soc. Am. Bull., 96,* 663–675, 1985.

Oberthür, T. and R. Saager, Silver and mercury in gold particles from the Proterozoic Witwatersrand placer deposits of South Africa: Metallogenic and geochemical implications, *Econ. Geol., 81,* 20–31, 1986.

Phillips, G. N., Metamorphism of the Witwatersrand gold fields: conditions during peak metamorphism, *J. Metamorphic Geol., 5,* 307–322, 1987.

Pullaiah, G., E. Irving, K. L. Buchan, and D. J. Dunlop, Magnetization changes caused by burial and uplift, *Earth Planet. Sci. Lett., 28,* 133–143, 1975.

Schweitzer, J., and A. Kröner, Geochemistry and petrogenesis of early Proterozoic intracratonic volcanic rocks of the Ventersdorp Supergroup, South Africa, *Chem. Geol., 51,* 265–288, 1985.

Steiger, R. H., and E. Jäger, Subcommission on geochronology: Convention on the use of decay constants in geo- and cosmochronology, *Earth Planet. Sci. Lett., 36,* 359–362, 1977.

Tankard, A. J., M. P. A. Jackson, K. A. Eriksson, D. K. Hobday, D. R. Hunter, and W. E. L. Minter, *Crustal Evolution of Southern Africa,* Springer-Verlag, New York, 1982.

N. Clauer, Centre de Sédimentologie et Géochimie de la Surface, CNRS, 1 rue Blessig, 67084, Strasbourg, France.

A. Kröner, Institut für Geowissenschaften, Johannes Gutenberg-Universität, Postfach 3980, Mainz, West Germany.

P. W. Layer, Department of Physics, University of Toronto, Toronto, Ontario, M5S 1A7, Canada.

M. McWilliams, Department of Geophysics, Stanford University, Stanford, California, 94025, USA.

(Received May 14, 1987;
revised October 13, 1987;
accepted October 20, 1987.)

21562202b, 1988, B3, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB093iB03p02191 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License